Game design document

Written 2026-10-12, after WP-C3. Revised the same day after your first read, and again after WP-C4, which built the four things you decided then: the board is seen from straight above, during a wave too (§3.1); walls are single tiles and no longer tetromino pieces (§4.1); the map has rocks (§3.2); and the board draws a route line for every kind of enemy you have met (§5.3, which also says where enemies walk). The board is 24 × 24 for now (§3.3). WP-C5 then built the five towers and four enemy types this file proposed, with new wave numbers (§4.2, §5.4, §6.1). After you played that, a trial of cards, tower levels and gear was built the same day (§4.4). No name has been chosen for the game yet (§12, question 12).

This file describes the game from the player's side: what is on the screen, what the player does, and what the finished game is meant to be. The other documents answer other questions: `CLAUDE.md` says what the thesis claims and what must never be broken, `ARCHITECTURE.md` how the code is built, `WORKPLAN.md` in what order. Where this file and `CLAUDE.md` disagree, `CLAUDE.md` wins.

Much of the finished game is not decided yet. So every part of this file carries one of four marks:

Mark	Meaning
BUILT	In the game today, and tested.
PLANNED	Decided and in the work plan. Not built.
PROPOSED	A suggestion made in this file, so that there is something concrete to look at. Not decided. Accept it, change it or drop it.
OPEN	Nobody has decided. §12 lists every open question, with who must answer it and by when.

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1. The game in short

A single-player maze tower defense on one square board, seen from straight above. Enemies come out of a spawn in one corner and head for your core in the opposite corner. You have one budget and spend it on two things:

- **walls** (single tiles that you draw onto the board), which bend the enemies' route;
- **towers** (bought from a shop that offers four at random), which shoot them.

The unusual rule: **nothing you build blocks the path.** Every wall and every tower can be chewed through. An enemy takes the cheapest way to the core, and "chew through this" is priced into that choice like any detour. So you never close a route; you make it long and expensive, and you put towers along it.

Waves come one after another, each stronger than the last. Between waves you build, buy and sell. The session ends when ten enemies have reached the core.

What the thesis adds. The game exists in two versions that differ in one thing only. In the *static* version every wave is the same kind of enemy, just more and tougher. In the *adaptive* version a wave director looks at what you built, picks the kind of wave that presses on that build's weak point, and tells you so in one line. Both versions send waves of the same total strength. The claim to be tested is that the adaptive version keeps a player rebuilding, where the static one lets them find one good maze and repeat it.

Design pillars

1. **You bend the path, you never close it.** A path to the core always exists.
2. **Two levers, one budget.** Towers are damage. Walls are time under fire. Walls alone kill nothing.
3. **Every wave asks a question about your build**, and says the question out loud.
4. **Fair.** Whatever the director sends, it has the same strength as the static wave of that number, and the director never touches your shop, your prices or your money.
5. **Readable.** The routes are drawn (§5.3). A tower shows its range before you pay for the tile. An enemy shows its health. Nothing important is hidden in a number.

2. One wave, from start to finish

```
BUILD PHASE      45 seconds before wave 1, 10 seconds after every wave
  │              draw walls · buy towers and place them · sell what did not work
  │              Start Wave skips the rest of the clock
  ▼
THE WAVE STARTS  the banner says what is coming
  │              enemies leave the spawn one after another
  │              towers fire · enemies walk the route, or chew through what is in the way
  │              you may still place walls and buy towers; you may not sell
  ▼
THE WAVE ENDS    when the last enemy has been killed or has reached the core
                 +20 budget · four new shop offers · the next build phase begins
```

All of this is BUILT, except the banner (PLANNED , WP14).

3. The board

BUILT (WP-C4). One map, 24 × 24 tiles, with rocks on it (§3.2). The spawn is near one corner (tile 2, 2) and the core near the opposite one (tile 21, 21). With nothing built the enemies' way is about 30 tiles long, which takes about three quarters of a minute at wave-1 speed. Any tile that is not a rock can be built on, except the spawn's and the core's.

3.1 Seen from straight above

BUILT (WP-C4; your decisions of 2026-10-12). The board lies flat on the screen and all of it is visible at once, as in the pictures of §5.3 and §9. There is no camera to turn or move, so four controls go away (§10).

The wave is watched in the same view as the build phase. One view to learn is less than two, and a participant has a single sitting to learn the game in. PROPOSED, not built: a flyer gets a shadow under it, so that "above the walls" can still be read from above.

3.2 Rocks

BUILT (WP-C4; your decision of 2026-10-12). An empty board of equal tiles gives a new player nothing to start from. So the map gets **rocks**: fixed tiles that belong to the map and not to the player.

- Nothing can be built on a rock. A rock cannot be sold and cannot be chewed.
- Enemies on foot cannot cross a rock. Flyers pass over it.
- The rocks are the same in every session, for every participant, in both versions of the game. They are drawn by hand and not rolled from the seed: the shop is one source of luck already, and the study has no use for a second.

Why rocks, and not walls that are already standing. A wall belongs to the player. It can be sold for budget, and it pays when it is chewed through, so a board that starts with walls starts with money lying on it. A rock is only ground.

What the rocks must not do is decide the maze. The study measures how a player's building changes from wave to wave. If the rocks leave one sensible line, every participant builds that line in both versions and there is nothing left to measure. So:

1. Few: about a tenth of the board. (The game's board has 68 rock tiles, 11.8 %.)
2. In separate groups, with more gaps between them than a player can afford to close. Which gaps to close is then the player's decision, and the waves give a reason to change it.
3. None close to the spawn or the core.
4. Always a way through on foot. The game checks this when it loads a map, and refuses a map whose rocks cut the spawn off (BUILT).

The game has a first layout, drawn by these four rules and checked with the path rule: on the empty board the walkers' way is about a tenth longer than the straight line. It is a first layout only. It is text in the scene (one line per row, `.` and `X`), so you can change it by playing; it is checked again in WP-C5. The pictures in §5.3 and §9.2 were drawn before it, on a larger example board.

Rocks take the place of the "seeded maze" that the schedule named for the second semester (question 11).

3.3 Size

BUILT as 24 × 24, which you chose "for now" on 2026-10-12 (question 19 stays open in that sense). The board was 38 × 38 before: a tile was small on the screen, one wall tile changed little, and the walk alone took over a minute a wave. Whether 24 is right is a thing to judge by playing and in the balance check (WP-C5); it also bears on how long a session is (question 1).

3.4 Spawn points

PLANNED **(WP9):** more than one spawn point, because one of the director's four wave types sends groups from different sides. Where the extra spawns go is OPEN (question 10).

4. What you build

4.1 Walls

BUILT (WP-C4; your decisions of 2026-10-12): **walls are no longer tetromino pieces.** They are single tiles:

- **A wall is one tile.** With the Wall tool, a click builds one tile; holding the button and dragging draws a line of them. Each tile is paid for as it appears, and the line stops when the budget runs out.
- **Every tile costs the same:** 1 budget (a placeholder, like every number here).
- A wall goes on any free tile: not on a rock, a tower, another wall, the spawn or the core. The tile under the mouse shows whether the spot is legal before you click. That preview and the click use one rule, so they can never disagree.
- **A wall is expensive ground, not an obstacle.** To an enemy, stepping onto a wall tile counts like walking 15 tiles. If the way round is longer than that, the enemy goes through: it stands on the tile and chews it (one enemy needs 6 seconds), the tile disappears, and every enemy's route is worked out again. You are paid 1 budget for a tile that is chewed through. With a tile priced at 1 that is the whole price back, which makes the balance question about it sharper (question 9).
- **Thickness is the first thing to watch.** A wall one tile thick costs a walker 15 tiles' worth to cross, and a sapper only 3. Two tiles thick costs 30 and 6.
- **Length is the second.** Walkers go round a wall only while going round is cheaper than chewing. Stretch the detour too far and they chew through the thinnest spot instead. (This happened while the picture in §5.3 was being made: with one baffle three tiles longer, the walkers went through the thin wall marked 1.)
- **Selling.** Between waves, with the Sell tool, a click sells a wall tile for half its price, and a drag sells every wall tile it passes over.
- There is no bag, no queue of next pieces, no rotation and no hold slot.

What the change does:

- **Less to learn.** Two controls and a row of the screen go away. Building is "draw where you want a wall".
- **For the study, one source of luck fewer.** Which piece came next nudged where a player built, in both versions. The measure of rebuilding counts built tiles by region of the board (`ARCHITECTURE.md` §5.9), so its definition does not change.
- **Smaller answers become possible.** A player can now meet a wave by adding two tiles, where a whole piece was the least before. The measure will see a small patch as a small change. Whether it still tells sessions apart is something to look at in the pilot, before the measure is frozen.
- **What is lost:** the small puzzle of fitting the piece you were dealt.

- **In the code** the rules did not change: a wall "piece" could already have any shape, one tile included. What changed is the list of shapes, the input, the screen, and every recorded session (they were recorded again).

Until WP-C4 the game had the tetromino version:

- Seven shapes of four tiles each (I, O, T, S, Z, J, L) from a shuffled bag. The current piece and the next three are shown. R rotates the piece; Shift swaps it with a hold slot.
- A piece costs 3 to 7 budget (I 4, O 7, T 5, S 3, Z 3, J 3, L 3) and goes anywhere its four tiles are free.
- A whole piece is sold at once for half its cost, or in proportion less if part of it has been chewed away.

4.2 Towers

BUILT, with a placeholder roster.

- A tower stands on one tile. It is expensive ground too (it counts like 20 tiles of walking), and an enemy that chews through it destroys it. That pays nothing.
- **The shop.** At the start of the game, and each time a wave ends, the shop shows four towers picked at random. Each offer can be bought once. Offers you did not buy stay through the next wave's fight and are replaced when that wave ends. The shop is open during a wave.
- **The shop is the same for everyone on the same seed.** What it offers depends on the seed and the wave number and on nothing else: not on what you built, not on what you bought, not on which version of the game is running.
- **How a tower fights.** Every tower fires on its own. It picks the enemy in range that is nearest to the core (measured along that enemy's own way there), hits it at once, and waits for its next shot. A splash tower also hits everything close to its target. A slowing tower makes its target walk slower for a moment.
- Between waves you can **sell** a tower for half of what you paid.

BUILT (WP-C5): the five towers this file proposed (question 2 is still yours). The numbers passed a sanity check and are not balanced:

Tower	Price	Damage type	What it does	Range	Hits flyers
Archer	10	physical	4 damage every 0.5 s to one enemy	3.5 tiles	yes
Ballista	16	physical	22 damage every 2 s to one enemy	5.5 tiles	yes
Cannon	18	fire	3 damage every 1.2 s to its target and everything within 0.75 tile	2.5 tiles	no
Brazier	12	fire	1.2 damage every 0.1 s to one enemy	1.5 tiles	no
Frost	12	frost	2 damage every 0.6 s; the target moves at half speed for 1.5 s	2.5 tiles	no

What each is for:

Tower	Strong against	Weak against
Archer	flyers, stragglers	a dense swarm
Ballista	tough enemies, diggers stuck at a wall	a dense swarm (its one heavy shot is wasted on a weak enemy)
Cannon	a swarm on a narrow route	flyers, enemies spread out
Brazier	a route that runs right past it	anything that goes round it or over it
Frost	makes every tower near it better	being alone

In the sanity check the four damage towers gave about the same damage for their price (within a fifth of each other). Frost cannot be judged that way, since what it does is slow. The numbers are tuned in the balance pass.

4.4 Cards, levels and gear

BUILT (2026-10-12), as a trial after your first play: the game was too rich after wave 3, one stacked chokepoint won it, and the random towers did not matter. Nothing here is balanced, and whether it stays is OPEN (question 20).

- **The build phase has no clock.** A wave starts when you press Start.
- **Money is tighter and no longer depends on kills.** You start with 50 and get 16 for every wave cleared. A kill pays nothing, and neither does a chewed wall. (This also closes question 6: income no longer depends on what a wave is made of.)
- **A shop slot holds a card,** and a card is one of three things:

Card	What it is
Tower	as before; about one in five is **rare** or **epic** and costs what the tower and its upgrades would
Coin	8 budget, free: click it to take it
Gear	a piece a tower wears, 8 budget: click the card, then a tower

- **A tower has three levels:** common, rare, epic. The Upgrade tool raises a tower one level for its card's price (half as much again for the second). A level changes what the tower does:

Tower	Rare	Epic
Archer	faster shots	hits two enemies
Ballista	two bolts	three bolts, +1 range
Cannon	bigger blast	+60% damage
Brazier	wider ring	burns the neighbours too
Frost	longer slow	slows an area

- **Gear.** A common tower has one gear slot, a rare or epic one has two.

Gear	What it does
Scope	+1 range
Whetstone	+35% damage
Sky-sight	the tower can hit flyers
Powder	+0.5 splash
Chill	slows what it hits

  Between waves the Move Gear tool takes a piece from one tower to another for nothing, so gear rewards re-arranging a defence. Selling a tower returns half of everything paid into it, its upgrades and the gear it wears included.
- **On the board** a rare tower stands on a blue plate and an epic one on a purple plate; each piece of gear is a small coloured dot on the tower.

Second round, the same day (also BUILT, also a trial):

- **You are told what comes next.** The next wave is decided when a build phase begins, from the board as it stood when the last wave ended. The top of the screen names it ("Next, wave 2: 22 sapper") and the route lines show where it will walk, so you build against it before you press Start. This settles question 7: the announcement comes at the start of the build phase, not as the wave starts.
- **Cards are a free pick.** After each wave three cards are shown (towers, sometimes rare or epic, and coins) and you take one. Before the first wave you take two. Towers come from nowhere else, so a defence grows by at most one tower a wave.
- **Gear is bought in a gear shop:** three pieces on sale for 8 budget each, a new stock every five waves. Gear is no longer a card.
- **Money** is 30 at the start and 10 a wave, and it goes on walls, upgrades and gear.

Known problem: the first cards decide too much. A start that offers two Frost towers and a coin cannot kill the first wave (two of ten seeds in the scripted runs).

Third round, 2026-10-13 (also BUILT, also a trial; borrowed from Teamfight Tactics):

- **Reroll.** A button under the cards buys a new set: 2 budget, then 3, then 4, back to 2 when the next wave's cards come. A pick you have not used carries over.
- **Three of a kind combine.** Three towers of one type and one level become one tower of the next level (three commons a rare, three rares an epic). The one you placed first stays where it is; the other two tiles become empty. Their gear moves onto it, and a piece that finds no slot is paid back. A card that would complete a set says so.
- **The Upgrade tool is gone.** Levels come from combining and from rare and epic cards.
- **A tower limit.** Four towers at the start, one more every three waves, shown as "towers 3 / 4". A tower that completes a set may always be placed.
- **Move.** Between waves a tower can be picked up and put on another free tile for nothing, with its level and its gear.
- **Gear is a card again** as well as being in the gear shop, and a roll shows four cards.

- **Three more towers,** eight in all: Sniper (physical, range 8, one heavy shot, ground only), Shard (frost, hits three enemies at once, ground only) and Flak (fire, splash, range 4.5, hits flyers). Their numbers are first guesses; Flak is at least as good as the Cannon on the ground and also hits flyers, which the next balance check has to settle.

What this risks for the study (to raise with your supervisor): a free move is a change to the layout that the primary measure does not count today (it counts tiles built), and a merge empties two tiles the player did not choose to empty. Both are written to telemetry (`moveTower`, `combine`), so the measure can be defined either way before it is frozen. Also: money put into levels is money not spent on placing or moving anything, and a tower with levels and gear is one a player is slower to sell. Both work against the rebuilding the study measures, in both versions. Upgrades and gear moves are in the build log and the telemetry, so they can be counted; the placement measure of §5.9 of the architecture does not see them.

4.3 How the two work together

A tower only earns its price while enemies are in its range. A wall puts no damage on anyone, but a route bent past the same towers twice doubles what those towers do. So the strong builds are mazes with towers at the bends, and every such build has a weak point somewhere: a wall one tile thick, a single spot that does all the killing, nothing that can shoot upward. Those weak points are what the adaptive version goes looking for.

5. Enemies

5.1 What an enemy is

BUILT An enemy has hit points, a walking speed, a digging speed, and one resistance per damage type (a resistance of 0.5 halves the damage of that type). It ends in one of two ways: **killed** (you are paid 0.2 budget) or **through** to the core (the core loses 1).

5.2 Three ways of moving

BUILT (WP-C2). This is what the director's wave types are made of.

Class	How it gets to the core	Digs	Which towers can hit it
Walker (ground)	Takes the cheapest way, counting a wall as 15 tiles of walking. Usually goes round.	Only when going round costs more.	All
Sapper	The same, but a wall counts as 3 tiles. Goes through walls that walkers go round.	Often, and its group digs faster.	All
Flyer	A straight line to the core, over everything.	Never.	Only towers that hit flyers

On the screen a walker is a small square, a sapper an orange disc and a flyer a blue diamond (WP-C4). Plain shapes: models are art, and art is the second semester's.

5.3 Where enemies walk FIGURE 1

BUILT The pink and the orange route in the picture were not drawn by hand: they are what the game's own path rule gives for that board (an example board, larger than the game's).

Nobody assigns a path to an enemy. There is one rule, and every enemy follows it on its own: *at every step, go to the neighbouring tile from which the rest of the way to the core is cheapest for my kind.* The way is priced in tiles of walking, and a built tile is priced as extra walking: 15 tiles for a walker, 3 for a sapper. A flyer does not use the rule at all. It flies a straight line.

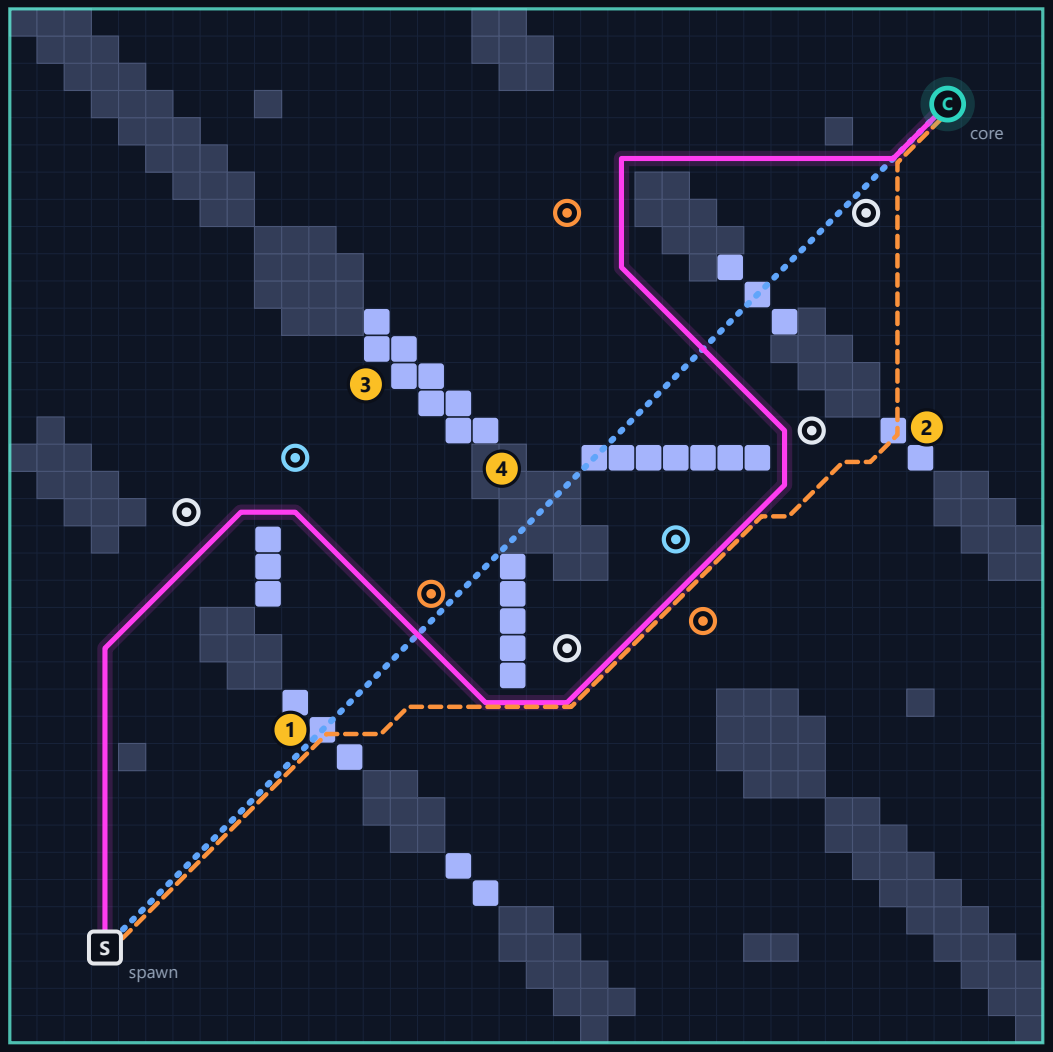
What follows from that:

- **Same kind, same entrance, same route.** All the walkers of a wave come out of the same spawn and walk the same line, one behind the other. They do not spread out, and they do not pick different gaps.
- **Different kinds, different routes, on the same board.** On the board of the picture:

Kind	Its way	Length	Towers it passes
Walker	round every wall	about 71 tiles	all 9
Sapper	through the two walls that are one tile thick	about 50 tiles	5
Flyer	a straight line over everything	44 tiles	1 (of the 4 that can hit flyers)

  So one maze is three different problems, depending on what arrives. That is what the director works with.
- **The director chooses the kind, never the path.** It decides what a wave is made of and, once there is more than one spawn (WP9), from which side it comes. After that each enemy finds its own way by the rule. The director cannot send an enemy to a particular tile, and no enemy is given a path of its own. Wherever an enemy goes, you could have read it off your own board.
- **A route can change in the middle of a wave.** When a wall or a tower is chewed through, or when you build, every route is worked out again and every enemy carries on from the tile it stands on.
- **A route is never closed.** Wall off every gap, and the wave chews through the cheapest spot.

FIGURE 1 **5.3 Where enemies walk**



Where enemies walk

One board, three kinds of enemy, three routes. Nobody steers an enemy: each one takes the cheapest way to the core for its own kind.

Walkers go round

To a walker, a wall tile counts like 15 tiles of walking. Going round is cheaper here, so they take the long way: about 71 tiles, past all nine towers. They chew nothing.

Sappers go through

To a sapper, a wall tile counts like 3 tiles. It cuts through the two walls that are one tile thick (1 and 2), leaves the thick wall (3) alone, and is there after about 50 tiles, past five towers.

Flyers go straight

A straight line over rocks (4), walls and towers: 44 tiles. Nothing you build bends it. Only towers that can hit flyers shoot at them; here one of those is in reach.

Same kind, same entrance, same route.

Enemies of one kind walk in single file along one line. The line is drawn on the board before the wave, so you can build against it.

The director chooses the kind, never the path.

It decides what comes (and later, from which side). After that every enemy finds its own way by the rule above. It cannot aim one at a tile.

A route can change in mid-wave.

When a wall is chewed through, or you build, every enemy on the board works out its way again from where it stands.

Dark blocks: rocks, part of the map. Light tiles: walls the player drew. Rings: towers. The pink and orange routes are what the game's own path rule gives for this board.

Where the three kinds of enemy walk on one board

What the screen shows. BUILT (WP-C4; your decision of 2026-10-12): in a build phase, one line for every kind of enemy you have met so far in this session; during a wave, the lines of the kinds in that wave. The lines are worked out again each time you build. Drawing all three from the first wave would be simpler, but in the static version it would warn of enemies that never come. With "the kinds you have met", the rule is the same in both versions and the static version keeps the one line it has today. Your supervisor should still see this before the pilot, because it is one of the things the screens of the two versions differ by.

5.4 Enemy types

BUILT (WP-C5), as this file proposed (question 3 is still yours): four named types, each a fixed set of numbers on top of a movement class.

Type	Class	Compared with the plain walker
Basic (the walker)	ground	the baseline
Swarm	ground	twice as many, four tenths of the hit points, 1.3 times as fast, half as far apart
Sapper	sapper	half as many, 1.5 times the hit points, 0.8 times as fast, digs four times as fast
Flyer	flying	half as many, half the hit points, 1.2 times as fast

Only the development planner sends anything but the basic type today. Which waves use resistances is OPEN (question 4).

6. Waves

6.1 The static version

BUILT This is the baseline condition of the study. Every wave is one stream of walkers from the one spawn, and every wave is a little worse than the one before:

Wave	Enemies	Hit points each	Speed (tiles per second)	Gap between enemies
1	40	8	0.63	0.40 s
5	60	22	0.70	0.36 s
10	85	39.5	0.79	0.31 s
15	110	57	0.89	0.26 s
20	135	74.5	0.98	0.21 s
25	160	92	1.08	0.20 s

(+5 enemies, +3.5 hit points, a little more speed and a slightly shorter gap each wave.) These numbers are from the sanity check of WP-C5 (`Results/2026-10-12_balance-v1`), where a scripted player that only buys towers loses around wave 11. Until then every wave was 100 or more enemies a fifth of a second apart, and a run ended in wave 1 or not at all. They are not balanced: that is the second semester's pass, against people.

6.2 The adaptive version

PLANNED (WP8 to WP11). Nothing of the director is built yet. What exists is a development planner that sends walkers, sappers, flyers and a mix in a fixed cycle, so that the three classes can be played.

What the director looks at. Once per wave, at the moment the build phase ends, it measures five things about your build:

Measure	In plain words
maze length	how much longer the route is than on an empty board
breach vulnerability	how much of the route depends on walls or towers that could be chewed through
chokepoint reliance	whether your damage is done in one spot or spread along the route
tower concentration	whether your towers stand together or apart
damage type mix	whether your damage is one type or several

Their exact formulas are OPEN (question 5).

What it can send. One of four wave types. The third column is what each is *meant* to press on. Which builds each one really moves a player to change is not a rule of the game: it is what the study measures.

Wave type	What arrives	Meant to press on
Breach the thin wall	sappers	a long maze held by thin walls
Swarm the chokepoint	many weak, fast enemies packed together	all the damage being done in one spot
Split groups	groups from more than one spawn	towers standing together in one place
Flying bypass	flyers	relying on the maze, with nothing that shoots upward

How many enemies of which type each wave type contains is OPEN (question 4).

What keeps it fair. These are fixed rules of the project (`CLAUDE.md` §4):

- **Same strength.** Every wave type costs exactly what the static wave of that number costs, on one shared price list for enemies. The director chooses the shape of a wave, never its size.
- **Nothing it cannot be answered with.** Flyers are not sent while you own no tower that can hit them and the shop is not offering one.
- **No repeats in disguise.** A wave too similar to the previous one is ruled out.
- **Hands off your side.** The director never changes the shop, a price, a tower or your budget. (Today one thing leaks through: you are paid per kill, so a wave of fewer, tougher enemies pays less. That has to be closed before the wave types are priced, question 6.)

How it chooses. It starts from written rules: each wave type gets a score for the build in front of it. On top of that it learns, during your session, which wave types actually made *you* rebuild, and nudges close calls that way. The nudge is small by construction: it can re-rank wave types that scored about the same, never overrule a clear one. The player sees none of these numbers.

What it is rewarded for is not beating you. It scores a wave highly when you changed your build afterwards, when your core was still standing, when the wave got close to the core, and when the wave was different from the last one. A wave that ends your session scores badly, because a player who is out builds nothing more.

The banner. As each wave starts, one line of text says what is coming and what in your build it is aimed at. The lines are written by hand, 20 of them (four wave types × five situations), and never generated. Sample wording, PROPOSED:

Wave type	Sample line
Breach the thin wall	"Sappers. They dig where your wall is one tile thick."
Swarm the chokepoint	"A swarm. Too many for one kill zone."
Split groups	"Two groups, two directions. Your towers are all in one place."
Flying bypass	"Flyers. Your maze does not reach the sky."

When the banner appears. The director has to see your finished build, so it chooses at the moment the build phase ends. The line therefore appears as the wave starts, not while you are building. You answer it during the wave (walls and towers can be placed in mid-wave) and in the next build phase. Whether that is enough warning to feel fair should be playtested (question 7). The route lines of §5.3 help here: from the moment the wave starts, the banner names a line that is drawn on the board, not something you have to imagine.

6.3 What is the same in both versions

The board and its rocks, the shop's offers (on the same seed), every price and reward, the strength of every wave, and the screen. Only the shape of the waves differs, and with it the text of the banner (question 7) and which route lines there are to draw (§5.3).

7. Money

BUILT One budget pays for walls and towers. All numbers are placeholders.

	Amount
At the start	30
After every wave	+10
An enemy killed	nothing (it was +0.2 until 2026-10-12)
A wall tile chewed through	nothing (it was +1)
A coin card	+8
An upgrade	the tower's price; half as much again for the second
A piece of gear	−8
A tower chewed through	nothing
A wall	today −3 to −7 for a piece of four tiles; planned −1 a tile
A tower	−10 to −18 (§4.2)
Selling a tower	half its price back
Selling a wall	half its cost back (today a whole piece at once, in proportion to the tiles still standing; planned tile by tile)
Rerolling the shop	not in the game (the rule exists and is switched off)

Towers are free picks now; thirty at the start is two upgrades, or an upgrade, a piece of gear and some walls.

8. Losing, winning, and a session

BUILT The core has 10 hit points and loses one for every enemy that reaches it. At zero the game is over and the screen shows "You reached Wave N" and a Restart button. There is no way to win: the waves go on until the core falls.

OPEN **(question 1): how long is a session?** This is the most important open question in this file. The study compares how a player builds over the waves they *completed* in each of their two sessions, cut to the shorter one. A participant who falls in wave 4 of one session leaves three waves to compare. So the study needs a stated length (the project's notes assume about 25 waves) and a game balanced so that an ordinary player gets most of the way there.

PLANNED **(WP14): a study session.** The participant types a code on a start screen. The code decides their two sessions: one static, one adaptive, on two different seeds, in an order that differs between four groups of participants. Nobody plays the same seed twice, so nobody can remember the shop.

OPEN **(question 8):** how a participant learns the game before their first session.

9. What the screen looks like

9.1 Today FIGURE 2

BUILT (WP-C4). The board from straight above, early in wave 1. Dark blocks: rocks. Light tiles: two walls drawn by drag. The two rings with a barrel: towers. The pink line: the walkers' route; the yellow streak at its start is the wave coming out of the spawn. A cannon from the shop is in the hand: its card is lit, and over the board it shows as a green tile with its range ring and a tooltip. Right: the shop's four cards, the two tools (Sell is greyed while a wave runs), and the key to the route lines. Top: the core's health and the wave. Bottom: Start, speed and Pause. It is functional, not styled: the text font, the colours and the size of the enemies are whatever came first.

9.2 The target: a build phase FIGURE 3

A mockup, on the board of the picture in §5.3. What in it is decided and what is not:

On the screen	Status
The board seen from straight above	BUILT
Walls as single tiles, drawn with the Wall tool	BUILT (§4.1)
Rocks	BUILT (§3.2); the layout in this picture is only an example
A route line for each kind of enemy you have met, with a key	BUILT (§5.3)
The shop as four cards; a tower picked from its card follows the mouse with its range ring	BUILT
Sell as a tool that shows what would be sold, and the refund	BUILT
Where things are on the screen, and how it all looks	PROPOSED; today's screen (§9.1) has the tools in the right-hand panel

- **The shop.** Click a card, and the tower follows the mouse with its **range ring**; click a tile to place it. Esc puts it down.
- **Wall and Sell are tools**, and one is on at a time. With Sell on, pointing at a tower or a wall tile shows the refund, and a click sells it.

9.3 The target: a wave, in the adaptive version FIGURE 4

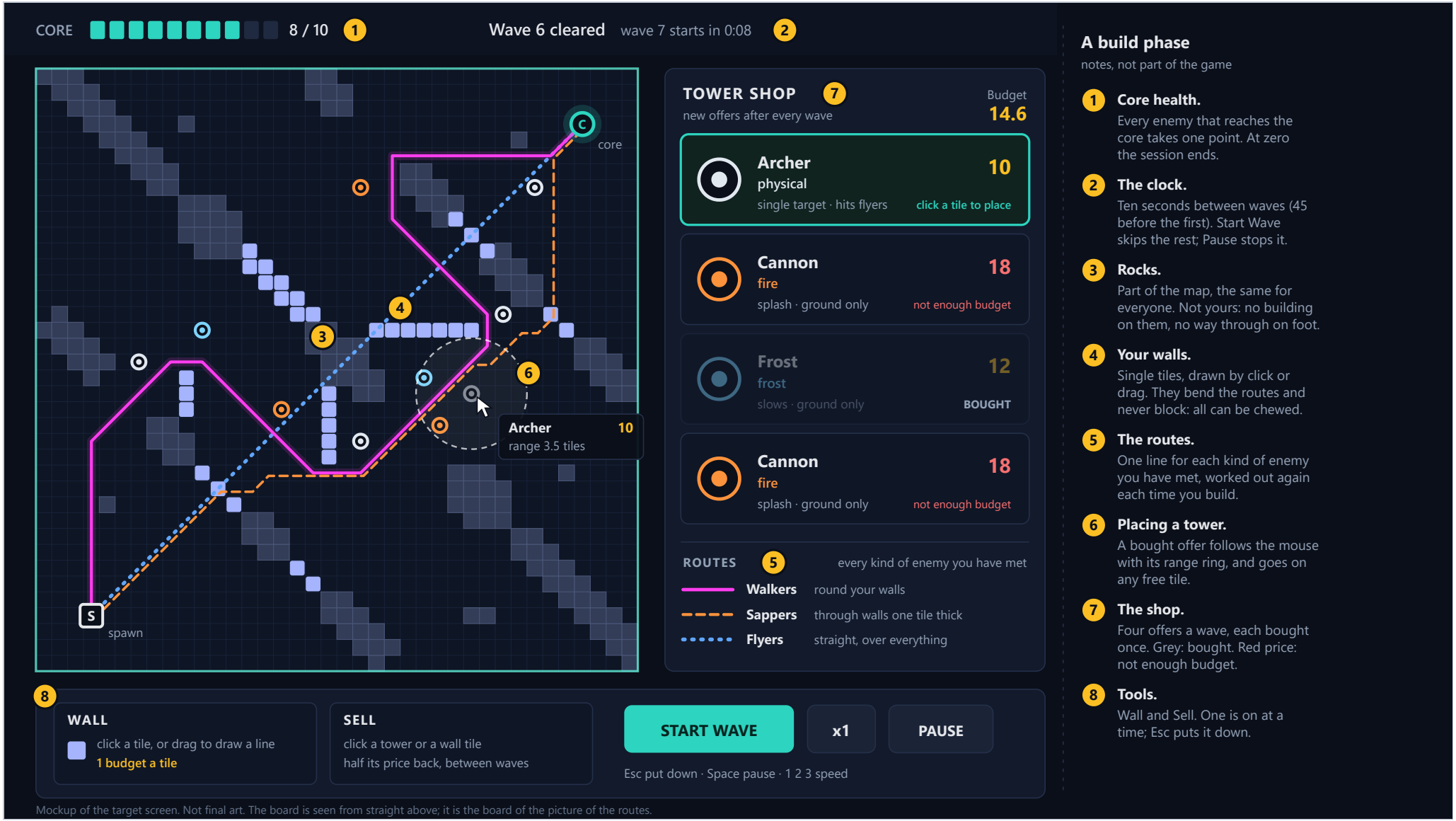
- The **banner** (PLANNED, WP14).
- Only the route of what is coming is drawn: here the sappers' (BUILT, §5.3). One of the two thin walls is already chewed through, and the first sappers are at the second.
- Towers that turn towards their target and fire a visible shot (BUILT: a line for a moment). The shot is only drawn; the damage is done the instant the tower fires.
- An enemy model per type. BUILT as plain shapes: a walker is a small square, a sapper an orange disc, a flyer a blue diamond. Models are art, and art is the second semester's.
- The Sell tool is visibly off until the wave is over (BUILT).

FIGURE 2 **9.1 Today**



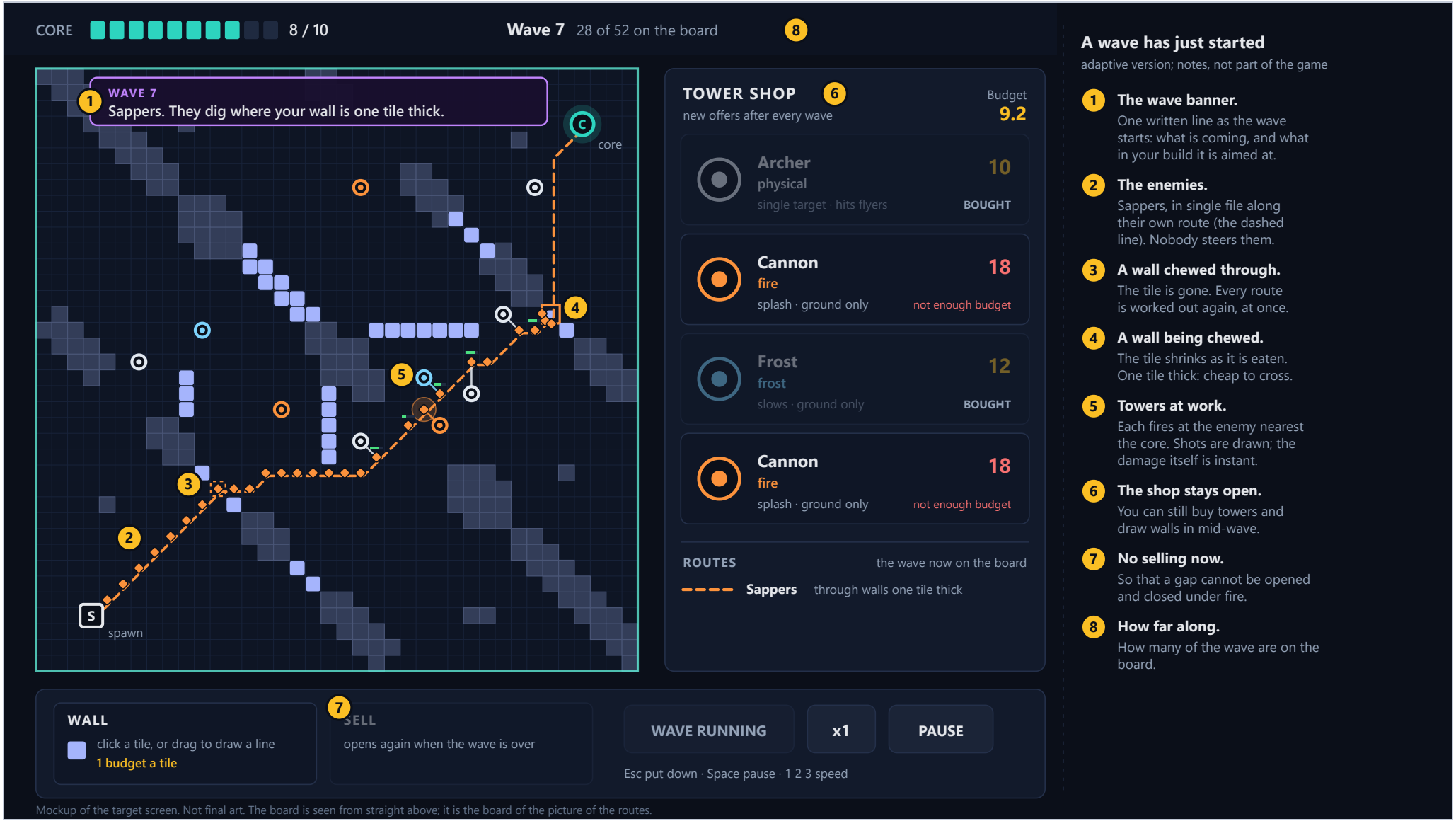
The game on 2026-10-12, after WP-C4

FIGURE 3 9.2 The target: a build phase



Mockup of a build phase

FIGURE 4 **9.3 The target: a wave, in the adaptive version**



Mockup of a wave that has just started

9.4 Other screens and overlays

Screen	Status
Game over: "You reached Wave N", Restart	BUILT
Flow-field view (key V): arrows, a heat map, or both, showing which way every tile leads	BUILT
Session start: participant code	PLANNED (WP14)
Director overlay (F9): why the director chose what it chose. For development; not in a participant's build	PLANNED (WP14)
Preview of how the routes would change, before a wall is paid for	PLANNED (second-semester UX pass)

10. Controls

BUILT (WP-C4).

Action	How
Build a wall	Wall tool (it is in the hand when the game starts): click a tile, or hold and drag to draw a line
Buy a tower	click its card, then click a tile; clicking the card again puts it back
Sell a tower or a wall	Sell tool, between waves: click it; drag to sell every wall tile the mouse passes over
Put the hand down	Esc, or the right mouse button
Start the wave early	Start button
Game speed	1, 2, 3
Pause (building is off while paused)	Space
Flow-field view	V, for development

For a participant that is seven controls where there were twelve. Gone with WP-C4: the keys that bought and sold (Z X C B, Delete), rotating and holding a piece (R, Shift), and everything that moved the camera.

11. Look and sound

Look. BUILT, as a placeholder: a dark board with a glowing edge, and everything on it a glowing cube. It is cheap, it reads well from above, and it is what the first pictures of the game look like. Rocks should look like ground and not like something built: darker and duller than a wall.

PLANNED **for the study build:** simple distinct shapes, so that a tower type and an enemy type can be told apart without reading (WP-C4), and then a pass over the whole screen in the second semester. Art beyond that is out of scope.

Sound. OPEN (question 13). There is none today and none in the plan.

12. Open questions

Each of these changes what the finished game is. "Suggestion" is this file's proposal, not a decision. Questions 15 to 19 came out of your first read on 2026-10-12. You answered 15 to 18 the same day; they are under the table.

#	Question	Suggestion	Who decides	Needed by
1	**How long is a session, and what ends it?**	A fixed length of 25 waves; the session still ends early if the core falls; balance so that an ordinary player gets past wave 15. Reaching the last wave is the win.	you and your supervisor	the balance check (WP-C5), firmly before the pilot
2	**Which towers?** (spec gap S12)	The five in §4.2 are built, because WP-C5 needed a roster. Still yours to change.	you	the balance pass
3	**Which enemy types?** (S12)	The four in §5.4 are built. Still yours to change.	you	WP9
4	**What exactly is in each of the four wave types,** and do they use resistances? (S10)	Take it from the proposal's §6.3, which is not in the repository yet.	you (add the proposal to `Docs/`)	WP9
5	**The exact formulas of the five measures.** (S9)	Take them from the proposal's §6.2. Starting definitions are in `ARCHITECTURE.md` §5.3.	you (add the proposal to `Docs/`)	WP8
6	**Income must not depend on the wave's shape.** (S14)	Closed in the trial of §4.4: a kill pays nothing and every cleared wave pays the same. If kill rewards come back, they must be priced by the shared list.	delegated	WP9
7	**The banner.** Does the static version show one? Is "as the wave starts" enough warning?	Both versions show a banner. The static one states the wave plainly ("Wave 7. 130 walkers."); the adaptive one also says why. Give every wave a few seconds between the banner and the first enemy, and playtest it.	you and your supervisor (it is part of the study design)	WP14
8	**How does a participant learn the game?**	A short practice run of three waves on a third seed, static, not analysed.	you and your supervisor	the pilot
9	**Balance questions:** does a chewed wall keep paying +1; the refund of a sale; rerolls.	Leave as they are until the balance check has numbers.	delegated	WP-C5, then the second-semester balance pass

#	Question	Suggestion	Who decides	Needed by
10	**Where do the extra spawn points go?** (S8)	The two corners the spawn and the core do not use, to be checked against the price list (from there the walk to the core is shorter).	delegated	WP9
11	**Is the "seeded maze" still wanted?**	No. The rocks of §3.2 do its job (the player does not start from an empty board), and they are fixed, so nothing new is random.	you	any time (the rocks are decided)
12	**A name for the game,** and the company and product name in the build settings (still `DefaultCompany/InternProj`). They decide where a build writes its data.	—	you	the first build that records data
13	**Sound.**	A handful of effects (place, shot, enemy through, wave start) in the UX pass, only if there is time. Not needed for the study.	you	the second-semester UX pass
14	**May a participant pause and change speed?** They can today.	Yes to both: pause is for thinking, and building is off while paused.	you and your supervisor	the pilot
20	**Do cards, levels and gear stay?** They were built as a trial (§4.4) and work against rebuilding in one way and for it in another.	Play it; then decide with the supervisor, since it bears on the measure.	you and your supervisor	before the pilot
19	**How big is the board?** It is 24 × 24 "for now" (your word, 2026-10-12).	Play on it; settle it with the balance check.	you	WP-C5

Answered

#	Question	Your answer	When
15	The walls, in detail	Yes, as in §4.1: single tiles drawn by click and drag, one price a tile, sold tile by tile.	2026-10-12
16	Rocks: yes or no, and where?	Yes, by the four rules of §3.2. A first layout is drawn in WP-C4 and you change it by playing.	2026-10-12
17	Is the wave watched from above too?	Yes: one view for the whole game.	2026-10-12
18	Which route lines are drawn?	One for each kind of enemy met so far in the session (§5.3). Still to be shown to your supervisor before the pilot, because the screens of the two versions differ by it.	2026-10-12

13. From today to the finished game

What each coming package changes for the person playing.

Package	What the player will notice
WP-C4 towers and shop on screen (**done**)	The board from above, with rocks on it; walls drawn tile by tile; a route line for every kind of enemy you have met. The shop as cards, a range ring before placing, a Sell tool with a preview, towers that turn and fire, a shape per enemy class.
WP-C5 roster and balance check (**done**)	Five towers and four enemy types; waves that start smaller and grow tougher.
WP7 the learner	Nothing. It runs without the game.
WP8 build profile	Nothing.
WP9 wave types	The four wave types exist; more than one spawn.
WP10, WP11 the director	Waves that answer the build.
WP14 director on screen, session start	The banner; the participant-code screen.
WP12, WP13 harness and gates	Nothing.
Second semester: UX pass, balance pass	Route previews, a clearer screen, a game that is fair to an ordinary player.
Build freeze	Nothing changes after this, except crash fixes.

The changes of 2026-10-12 were built with WP-C4, on the day they were decided. Nobody has played the result by hand yet: that is the next thing to do, before WP-C5 sets numbers on top of it.

14. What this game is not

- **Not a game that gets easier or harder to suit you.** Difficulty is held equal on purpose; only the shape of the waves changes.
- **Not a game where walls block.** That was the original design and was removed; it is what makes "a path always exists" true.
- **Not a puzzle of fitting shapes.** That was the tetromino version of the walls (§4.1).
- **No generated text or behaviour.** Every banner line is written by hand, and every run can be replayed exactly.
- **Not in the plan:** more than one map, anything carried over from one session to the next, more than one player. (Tower upgrades were on this list until 2026-10-12.)